ASSIGNMENT

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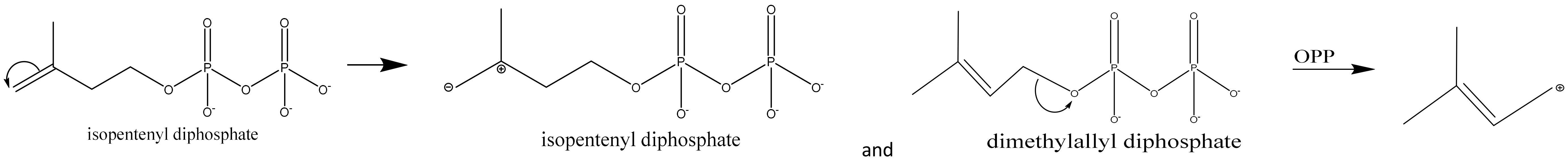
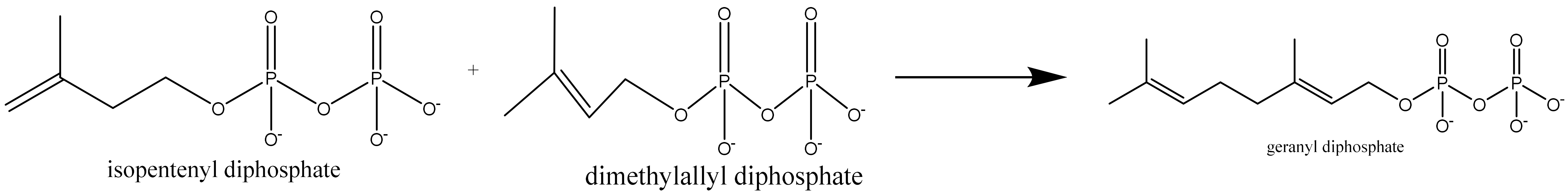
MATRI NUMBER	**F/HD/24/3710024**
COURSE TITLE	**NATURAL PRODUCT CHEMISTRY**
COURSE CODE	**STC 432**

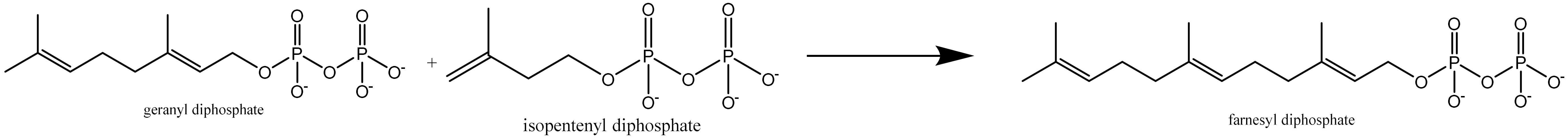
1. Using equations and mechanism only, ➔ (i) Geraniol, (ii) Farnesol, (iii) alpha-terpineol, (iv) Limonene

Answer:

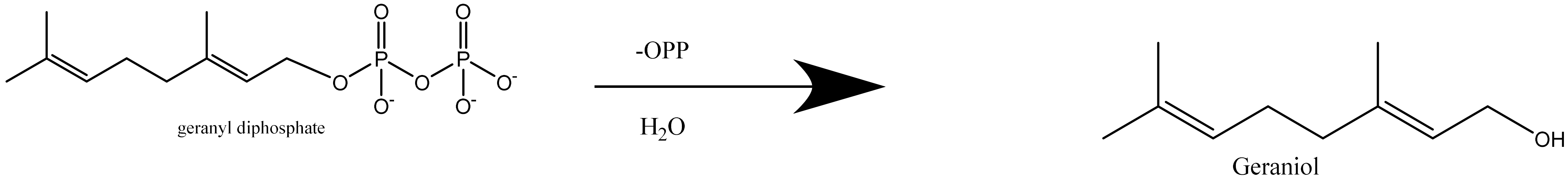
Starting materials are, dimethylallyl diphosphate and isopentenyl diphosphate

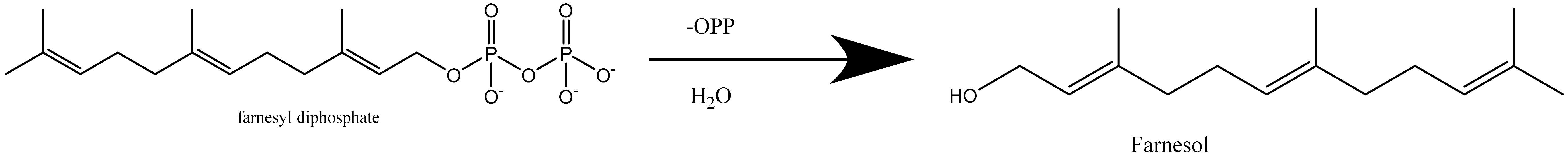




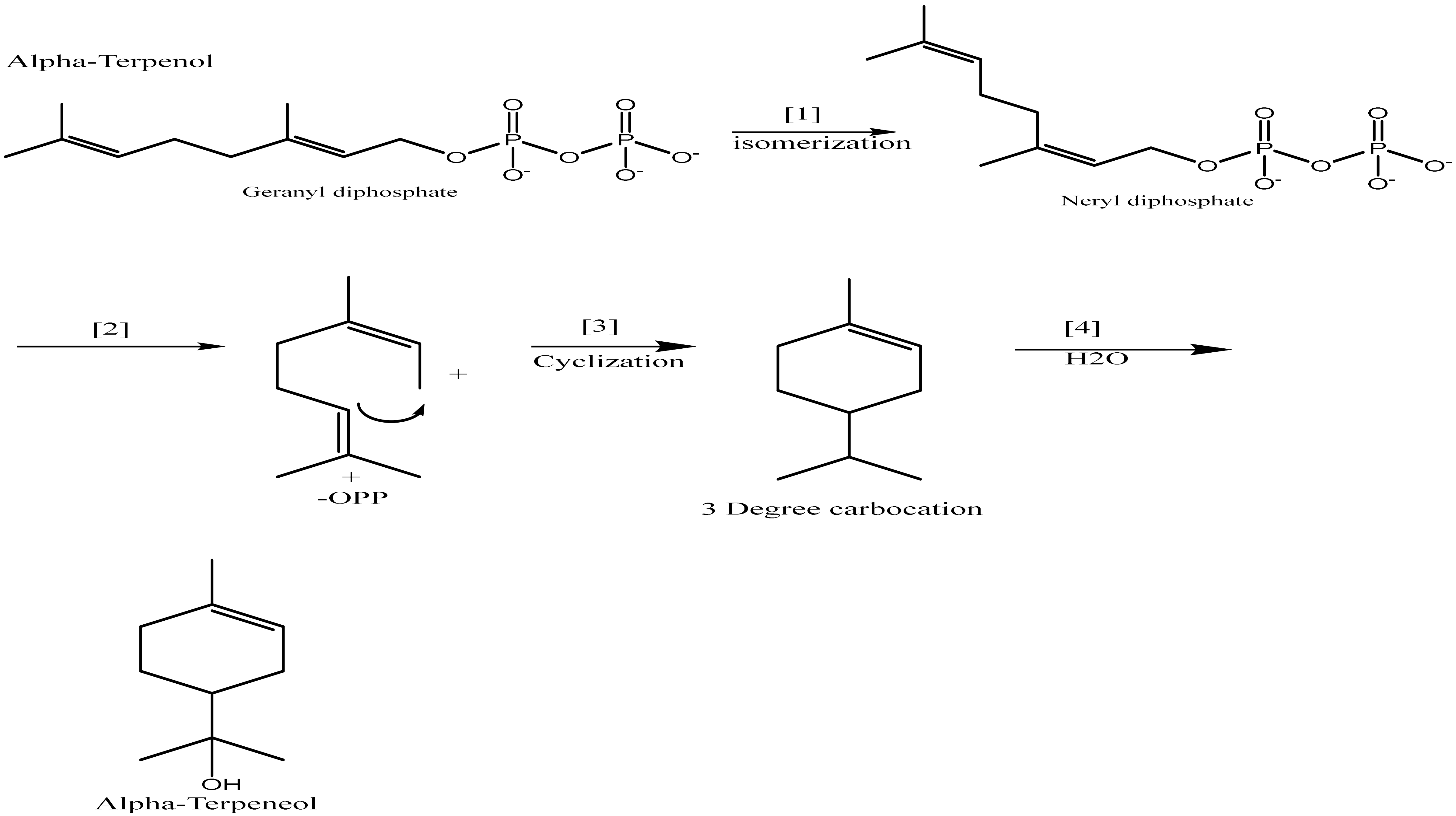


To synthesize Geraniol and Farnesol, we hydrate farnesyl diphophate and geranyl diphophate respectively

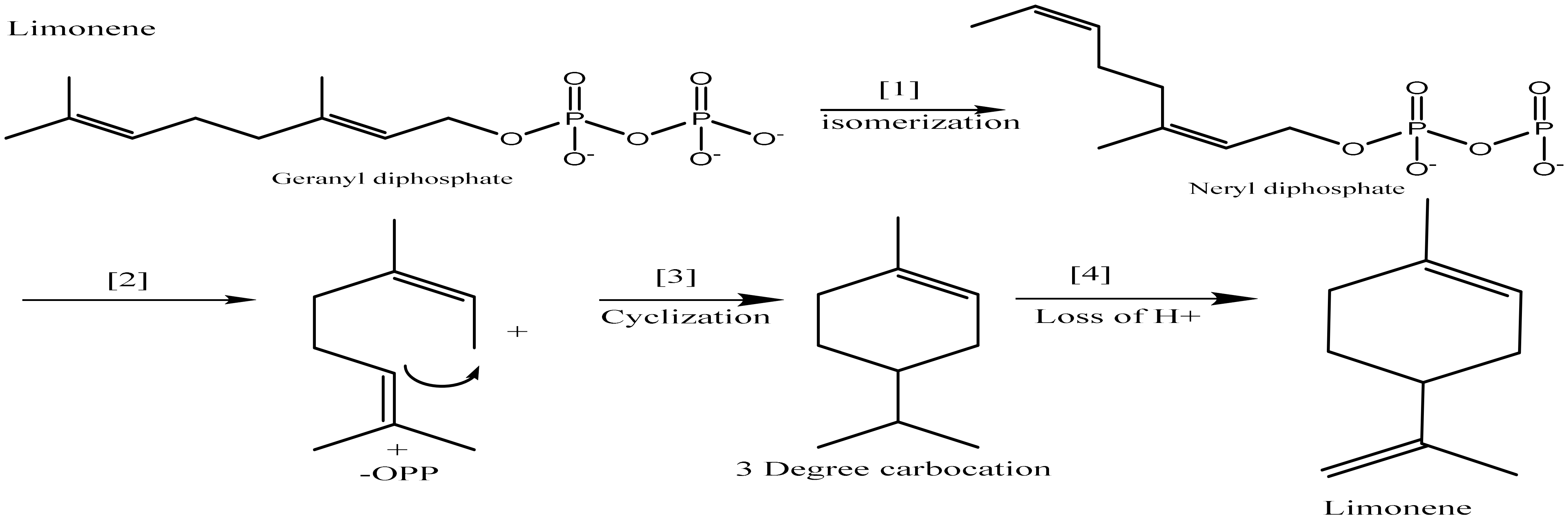




Alpha-Terpenol



Limonene



2a. What are steroids?

Answer:

Steroids are a group of tetracyclic lipid which are biologically active. They are widely present in plants and animals, They are derivatives of tetracyclic triterpenes.

In tetracyclic skeleton of steroids, the rings are numbered as A, B, C, and D. Cholesterol is one of the most important naturally occuring steroids which acts as a precusor for a number of other steroids.

2b. Discuss and categorize steroids.

Answer:

steriods are classified into five major categories:

i. Sterols:These are categorze by the present of an alcoholic group and are found in plant oil and animal fat

ii. Sex hormones: Unlike hormones sex hormones are not secreted by ductless gland, but are produced in gonads (testes of male and ovaries of female).

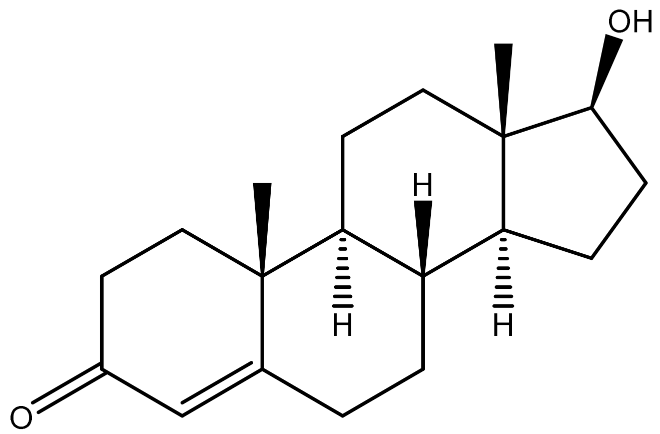
iii. Cadiac Glycosides: These Steriods are extracted from plant and are used for cadiac therapy for example;Digitoxigenin

iv. Bile Acids: These are organic compound present in the bile.

v. Sapogenins: These are present in the plant glycoside and their colloidal solution in water form foam similar to soap.

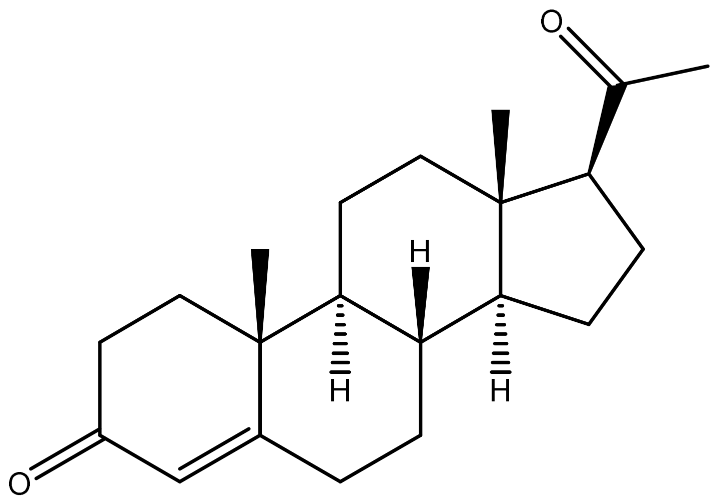
2c. Draw the structures of the following steroids:

(i) Testosterone



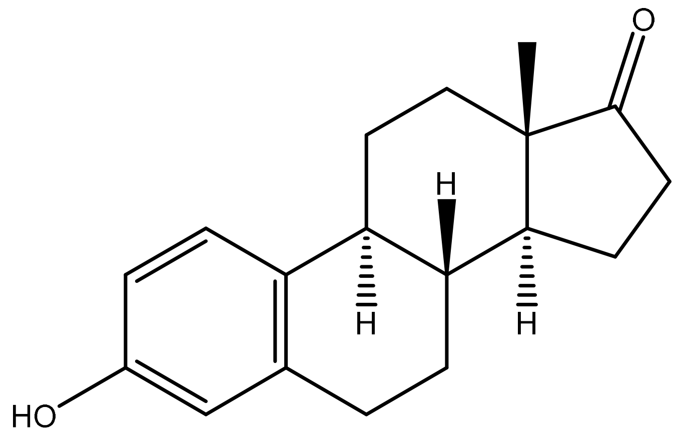
testosterone

(ii) Progesterone



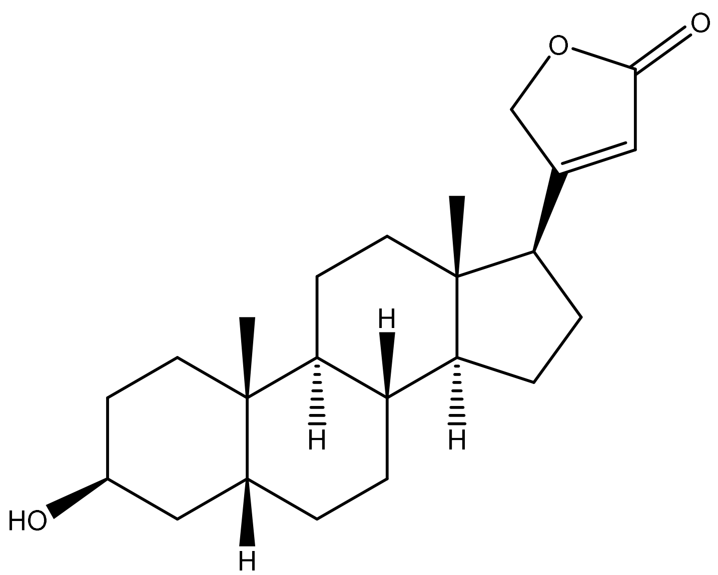
progesterone

(iii) Oestrone



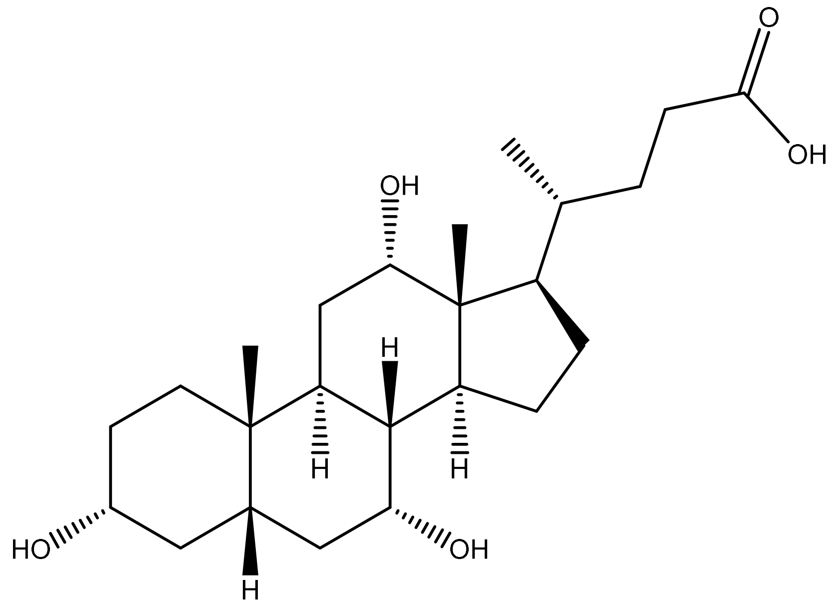
oestrone

(iv) Digitoxigenin



digitoxigenin

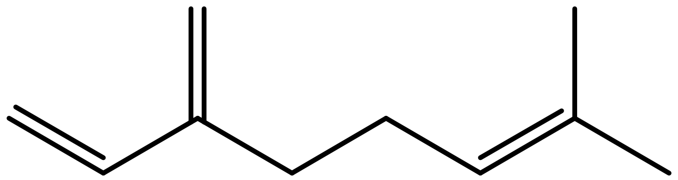
(v) Cholic acid



cholic acid

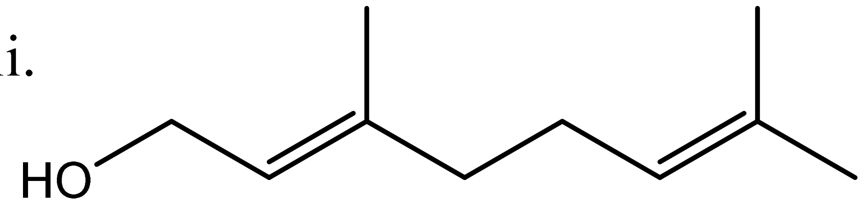
3a. Mention five monoterpenes and their source.

i.



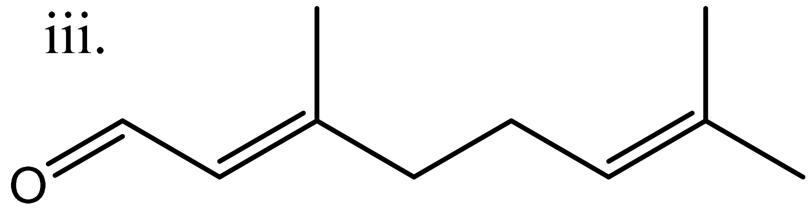
myrcene
bay oil

ii.



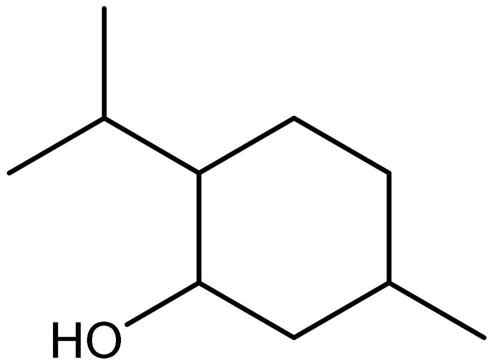
geraniol
rose oil

iii.



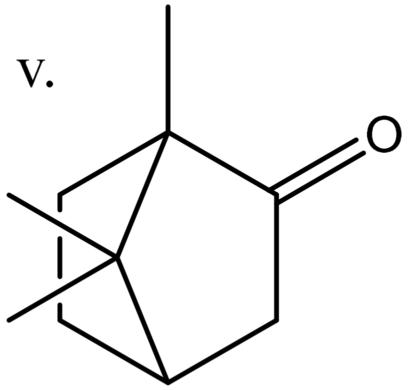
citral
oil of loemon grass

iv.



menthol
pepper mint

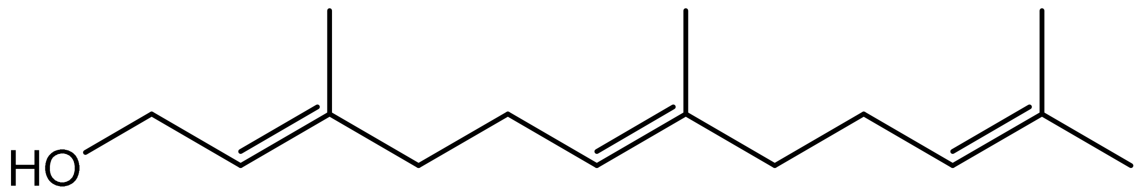
v.



camphor
camphor tree

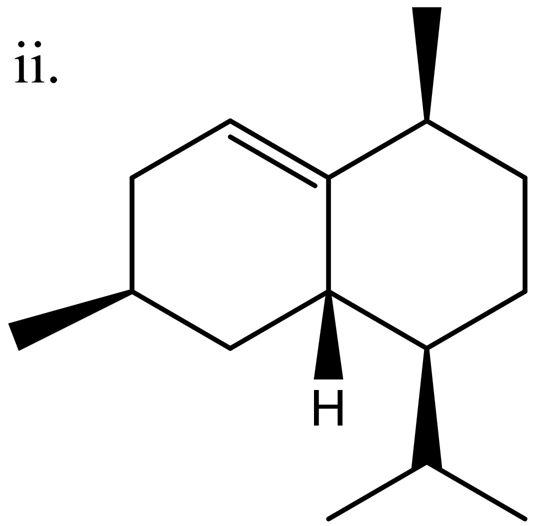
3b. Mention two sesquiterpenes and give its source.

i.



farnesol
cironella oil used in perfumes

ii.



cadinene

Juniper and cedar oil

3c. Discuss and classify the various classes of terpenes.

Classes of Terpenes

Monoterpenes: The simplest terpenes, composed of 10 carbon atoms formed from two combined isoprene units (dimers). They are classified based on their ring structures into acyclic, monocyclic, and bicyclic monoterpenes.

Sesquiterpenes: Contain 15 carbon atoms built from three isoprene units. They exist in various structural forms, including acyclic, monocyclic, bicyclic, and polycyclic ring systems.

Higher Terpenes:

Diterpenes (20 Carbons): Include compounds like phytol (an alicyclic alcohol component of chlorophyll) and Vitamin A (a monocyclic diterpene).

Triterpenes (30 Carbons): Exemplified by squalene, a hydrocarbon abundant in shark liver oils.

Tetraterpenes (40 Carbons): Consist of plant pigments known as carotenoids.

4. Outline the biosynthesis of cholesterol with equations only.